

Michel Zeller

SOFTWARE ENGINEER · COMPUTER VISION & MACHINE LEARNING

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About

I thrive on tackling complex problems from first principles, whether it's 3D reconstruction or building AI pipelines from scratch. What drives me is diving into new technologies and domains—from neural rendering to biomechanical analysis—and seeing my work make a real impact. I work best when I can take ownership of challenging projects while continuously learning and growing along the way.

Employment

CrossPath Labs

Zurich, Switzerland (Remote)

ML ENGINEER (FREELANCE)

May 2025 – Aug. 2025

- Designed and implemented the AI pipeline for the *eigen-running* platform, from data loading and biomechanical analysis using Vision Transformers to model evaluation and deployment using Docker on Azure Cloud
- Developed scalable API services with FastAPI to run the ML pipeline, powering the platform's first MVP

meshcapade

Zurich, Switzerland (Remote)

ML ENGINEER (INTERN)

Oct. 2024 – Mar. 2025

- Expanded MSc thesis project to target publication at top-tier CV conferences (CVPR/ICCV), focusing on 3D reconstruction and pose estimation
- Developed a method for understanding human-object interactions in 3D, outperforming state-of-the-art methods in accuracy and efficiency through novel neural rendering techniques (🧠human-object-interactions)
- Worked on λ 's large compute clusters to optimize the training and inference of deep learning models, leveraging distributed computing frameworks like PyTorch Lightning and AWS S3 buckets
- Onboarded to meshcapade's Rust-based 3D viewer codebase during a week-long hackathon, implementing the 'View Camera Frustum' feature while navigating unfamiliar architecture and collaborating with the team
- Participated in weekly ML team-meetings, sharing progress and exchanging feedback with a diverse research team

LOGIBLOX AG

Zurich, Switzerland

SOFTWARE ENGINEER (FULL-STACK)

Apr. 2022 – Dec. 2023

- Developed backend modules (graph compiler, AI components, database) and later Angular UI features for a no-code BI platform
- Delivered full-stack solutions with Python/Flask, SQL, TypeScript/Angular, and Google Cloud services
- Collaborated directly with customers and cross-functional teams in BI/Data Science projects, from requirements to onboarding

MeteoSwiss, Dep. Analysis & Numerical Predictions

Zurich, Switzerland

SOFTWARE ENGINEER (CIVIL SERVICE)

Sept. 2021 – Mar. 2022

- Migrated legacy visualization workflows from IDL to Python, eliminating yearly licensing costs
- Developed CLI tools for visualizing global air-trajectory data, still in operational use and further maintained at 🐍pytrajplot

Education

ETH Zurich, D-MAVT

Zurich, Switzerland

MASTER OF SCIENCE IN MECHANICAL ENGINEERING

Sept. 2022 – Sept. 2024

- Research-centered program with a focus on Deep Learning for 3D/Computer Vision and Robotics (5.61/6)
- Final Thesis: "Reconstructing Hand-Object Interactions in 3D from Monocular Video with 3D Gaussian Splatting"

ETH Zurich, D-MAVT

Zurich, Switzerland

BACHELOR OF SCIENCE IN MECHANICAL ENGINEERING

Sept. 2017 – Sept. 2021

- Strong STEM foundations & trained in parametrizing/solving problems. Focused on Energy, Flows & Processes.
- Final Thesis: "Drone Tracking in Challenging Conditions"

Skills

Programming

Python, C++, TypeScript, SQL, HTML/CSS

Machine Learning & Vision

PyTorch, OpenCV, Neural Rendering, SMPL, OpenPose, Scipy

Full-Stack & Cloud

Angular, Flask, Firebase, REST APIs, Linux, Git, Docker, Google Cloud, AWS, Azure

Languages

Swiss German (native), English (C2), French (reading/writing)